

**SECRETARÍA DE EDUCACIÓN PÚBLICA
TECNOLÓGICO NACIONAL DE MÉXICO
INSTITUTO TECNOLÓGICO DE VERACRUZ**

ASIGNATURA: Administración de bases de datos

GRUPO: 6J3

ACTIVIDAD: Practica 1 y 2

ALUMNO:
Chagala Cobarruvias Suany

NUMERO DE CONTROL:
22020749

SEMESTRE: AGOSTO – DICIEMBRE 2026

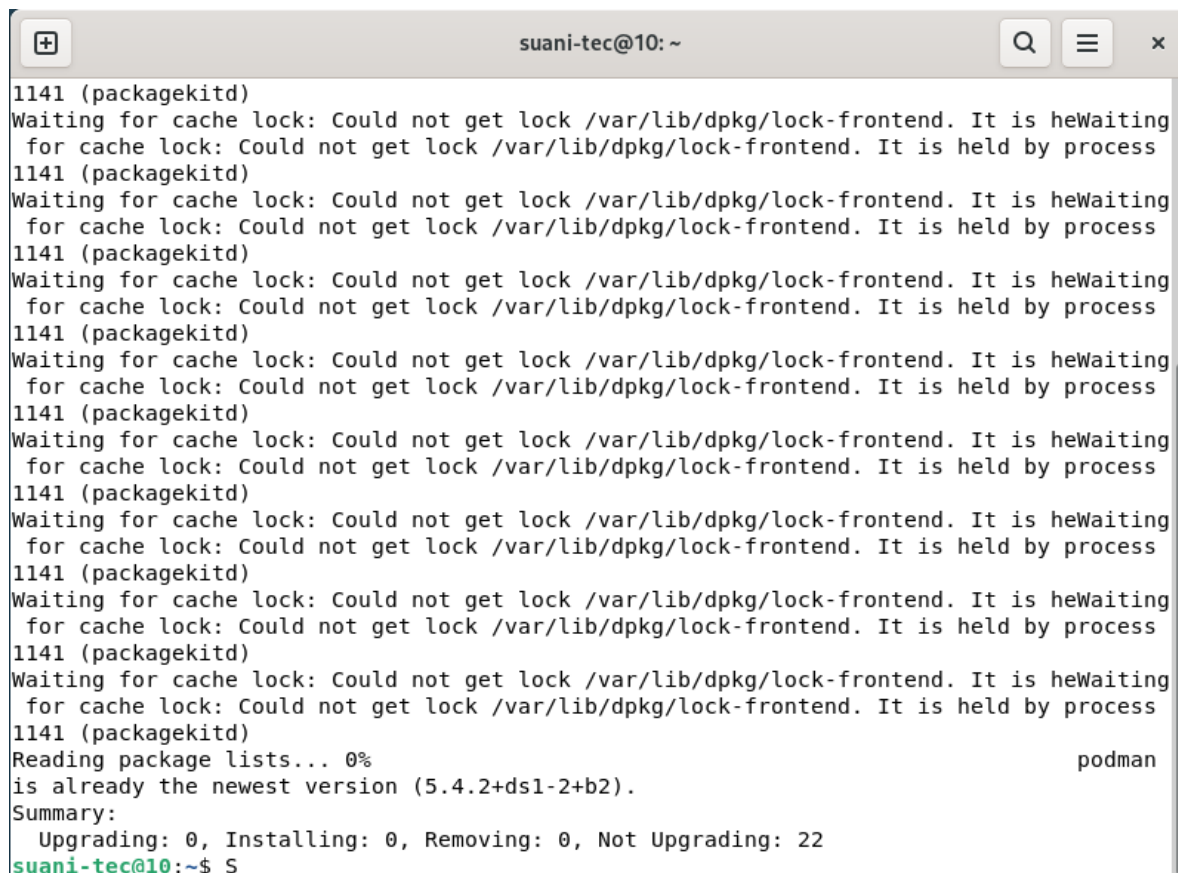
1. Verificación del Sistema Operativo y Entorno Virtual

```
suani-tec@10: ~  
suani-tec@10:~$ cat /etc/debian_version  
13.7  
suani-tec@10:~$ uname -r  
6.12.107+deb13-amd64  
suani-tec@10:~$ lsmod | grep -i vbox  
vboxguest          524288  2  
suani-tec@10:~$ hostnamectl  
Static hostname: 10  
Icon name: computer-vm  
Chassis: vm  
Machine ID: a50136154d2d4c518586e227897f56c9  
Boot ID: 0925d3d2e2f440cca8d85410c5e80a5e  
Virtualization: oracle  
Operating System: Debian GNU/Linux 13 (trixie)  
Kernel: Linux 6.12.107+deb13-amd64  
Architecture: x86-64  
Hardware Vendor: innotek GmbH  
Hardware Model: VirtualBox  
Firmware Version: VirtualBox  
Firmware Date: Fri 2006-12-01  
Firmware Age: 19y 9month 4w 2d  
suani-tec@10:~$ cat /etc/os-release  
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"  
NAME="Debian GNU/Linux"  
VERSION_ID="13"  
VERSION="13 (trixie)"  
VERSION_CODENAME=trixie  
DEBIAN_VERSION_FULL=13.7  
ID=debian  
HOME_URL="https://www.debian.org/"  
SUPPORT_URL="https://www.debian.org/support"  
BUG_REPORT_URL="https://bugs.debian.org/"  
suani-tec@10:~$ uname -a  
Linux 10 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64 GNU/Linux  
suani-tec@10:~$
```

2. Actualización del Sistema e Instalación de Podman

Actualizaste la lista de paquetes del sistema e intentaste instalar Podman

```
suani-tec@10:~$ sudo apt update && sudo apt install -y podman
Hit:1 http://deb.debian.org/debian trixie InRelease
Get:2 http://security.debian.org/debian-security trixie-security InRelease [43.4 kB]
Hit:3 http://deb.debian.org/debian trixie-updates InRelease
Fetched 43.4 kB in 1s (39.2 kB/s)
22 packages can be upgraded. Run 'apt list --upgradable' to see them.
podman is already the newest version (5.4.2+ds1-2+b2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 22
suani-tec@10:~$
```



```
suani-tec@10: ~
1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
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Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 1141 (packagekitd)
Reading package lists... 0%
is already the newest version (5.4.2+ds1-2+b2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 22
suani-tec@10:~$ S
```



3. Despliegue del Contenedor de Base de Datos MySQL

Descargue la imagen oficial de MySQL 8.4 desde el registro de Docker Hub y un contenedor con el nombre mysql

```
suani-tec@10:~$ podman pull docker.io/library/mysql:8.4
Trying to pull docker.io/library/mysql:8.4...
Getting image source signatures
Copying blob ff557ec8ac73 done |
Copying blob ff557ec8ac73 done |
Copying blob ff557ec8ac73 done |
Copying blob ff557ec8ac73 done |
Copying blob ff557ec8ac73 done |
Copying blob 0da02031b810 done |
Copying blob b23c4a59849a done |
Copying blob e49eff447cf4 done |
Copying blob 442cf3908783 done |
Copying blob 9056f6ecb394 done |
Copying blob 1c8d0b814939 done |
Copying blob 3a129719b3ee done |
Copying blob ea066d40f8be done |
Copying blob 1be21a6c94a6 done |
Copying config 8d132912e9 done |
Writing manifest to image destination
8d132912e9d3c985237567595ec2aeae04936044eb1f9336e8e4da2326238b84
suani-tec@10:~$
suani-tec@10:~$
suani-tec@10:~$
suani-tec@10:~$
suani-tec@10:~$ podman images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
docker.io/library/mysql 8.4                8d132912e9d3       43 hours ago       832 MB
suani-tec@10:~$ S
```

```
suani-tec@10:~$ podman create --name mysql -e MYSQL_ROOT_PASSWORD=Tec14 -p 3307:3306 do
cker.io/library/mysql:8.4
72180609e7447142628f9465b2e9d44744eb622bac8712fa7d9d1d657f3f1a
suani-tec@10:~$ podman start mysql
mysql
suani-tec@10:~$ podman ps
CONTAINER ID   IMAGE                                COMMAND              CREATED        STATUS        PORTS
72180609e744  docker.io/library/mysql:8.4         mysqld              59 seconds ago Up 21 seconds 0
.0.0.0:3307->3306/tcp, 33060/tcp mysql
suani-tec@10:~$
```

4. Despliegue del Contenedor de Base de Datos MongoDB

Descargue la imagen oficial de MongoDB y el contenedor de MongoDB configurando

```
suani-tec@10:~$ podman rm -f mysql
mysql
suani-tec@10:~$ podman create --name mysql -e MYSQL_ROOT_PASSWORD=Tec14 -p 3307:3306 docker.io/library/mysql:8.4
9b6f142226e2065aaa26a8e40ab177ef8f0f5fadc32494bb1a74d4b2240d888b
suani-tec@10:~$ podman start mysql
mysql
suani-tec@10:~$ podman ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
9b6f142226e2	docker.io/library/mysql:8.4	mysql	mysqld	About a minute ago	Up 13 seconds	0.0.0.0:3307->3306/tcp, 33060/tcp

```
suani-tec@10:~$
```

```
suani-tec@10:~$ podman pull docker.io/library/mongo:8.0
Trying to pull docker.io/library/mongo:8.0...
Getting image source signatures
Copying blob 051f5efcc90f done
Copying blob edd1ed89f0d4 done
Copying blob adf7abd1fe48 done
Copying blob 5e68fb960432 done
Copying blob 3bc1468bbf2a done
Copying blob 8d07092de802 done
Copying blob 730272832256 done
Copying blob 1aea056e9ccb done
Copying config 70ca9c7181 done
Writing manifest to image destination
70ca9c71813295005776e37005b0150235f84cf002588d2484d8f93747c401a8
suani-tec@10:~$ podman create --name mongo -e MONGO_INITDB_ROOT_USERNAME=admin -e MONGO_INITDB_ROOT_PASSWORD=Tec14 -p 27017:27017 docker.io/library/mongo:8.0
5285f23dbe468218c1484f67685abd554366711ee02cd4dd01292341bc80434a
suani-tec@10:~$
suani-tec@10:~$ podman start mongo
mongo
suani-tec@10:~$ ^[[200~podman ps~
bash: '$\E[200~podman': command not found
suani-tec@10:~$ podman ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
9b6f142226e2	docker.io/library/mysql:8.4	mysql	mysqld	9 minutes ago	Up 8 minutes	0.0.0.0:3307->3306/tcp, 33060/tcp
5285f23dbe46	docker.io/library/mongo:8.0	mongo	mongod	3 minutes ago	Up 35 seconds	0.0.0.0:27017->27017/tcp

```
suani-tec@10:~$
```

5. Copia e Importación de un Archivo SQL al Contenedor MySQL
Te desplazaste a la carpeta de descargas de tu máquina virtual.
Ejecute sesión interactiva para ingresar a la consola de comandos del servidor MySQL dentro del contenedor usando el usuario.

```
suani-tec@10:~$ podman cp ~/Downloads/tienda.sql mysql:/tmp/tienda.sql
suani-tec@10:~$ cd Downloads
suani-tec@10:~/Downloads$ podman cp tienda.sql mysql:/tmp/tienda.sql
suani-tec@10:~/Downloads$ podman exec -it mysql mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```